

AI Face Attendance System - Complete Engineering Notes

1. System Architecture & Flow

Pipeline: OpenCV Frame Capture → MTCNN Face Detection & Alignment → InceptionResnetV1 (512-d Embedding) → Typesense (Cosine Distance Search) → MySQL (Session Tracking) → FastAPI /attendance/logs.

2. Docker Infrastructure & Configuration

MySQL 8.0: Port 3306 | User: ai_attendance | DB: Ai_attendance
Typesense 27.1: Host Port 8109 (Mapped to Container 8108) | API Key: xyz123

3. Essential Terminal Commands Master Table

Purpose	Terminal Command
Free Port 8108	sudo fuser -k 8108/tcp
Kill Port PID	sudo kill -9 \$(sudo lsof -t -i :8108)
Check Container Key	docker exec -it attendance_typesense env grep TYPESENSE_API_KEY
Reset Volumes	docker compose down -v
Clean Stuck Containers	docker rm -f attendance_typesense attendance_mysql
Start Containers	docker compose up -d --remove-orphans
Run Server	uvicorn main:app --host 0.0.0.0 --port 8000 --reload
Query DB Directly	docker exec -it attendance_mysql mysql -u ai_attendance -p2003 -e "USE Ai_attendance; SELECT * FROM attendance_logs"
Clear DB Records	docker exec -it attendance_mysql mysql -u ai_attendance -p2003 -e "USE Ai_attendance; TRUNCATE TABLE attendance_logs"

4. Solved Issues & Key Insights

- 1. Port 8108 Conflict:** Resolved by shifting external bind to 8109 in compose and .env.
- 2. 401 Forbidden Typesense:** Resolved by removing stale volume caching via down -v.
- 3. Table Missing 1146:** Automated table bootstrap via init_db() hook inside FastAPI lifespan.
- 4. Live Verification:** Records accessible via Docker MySQL CLI or http://localhost:8000/attendance/logs.